

Chapter

Climate

1. What causes snowfall in Kashmir during winter? MAIN 2024

- (a) Tropical cyclone
- (b) Northeast Monsoon wind
- (c) Southwest Monsoon wind
- (d) Temperate cyclone

2. Which of the following area receives rain from the North East Monsoon? MAIN 2023

- (a) Konkan coast
- (b) Ganga basin
- (c) Coromandel coast
- (d) Malabar coast

3. The northern half of India lies in: SQP2023

- (a) Torrid Zone
- (b) Frigid Zone
- (c) Temperate Zone
- (d) Equatorial Zone

4. What is monsoon ? COMP 2024

5. Why does the Monsoon wind change its direction ? MAIN 2021

6. Name two months associated with retreating monsoons. MAIN 2021

7. Why does Patna get more rain than Agra during the summer monsoon season ? MAIN 2022

- (a) Patna lies on windward side.
- (b) Patna is on western side, so Arabian Sea branch brings more rain to it than to Agra.

(c) Patna is on eastern side, so Bay of Bengal branch brings more rain to it than to Agra.

(d) Patna gets rain throughout the year.

8. Name any two states with annual rainfall of less than 50 cm. MAIN 2023

9. Name any two driest stations of India. SQP 2024

10. Name three most rainy months of India as a whole. MAIN 2021

11. How is average rainfall calculated ? MAIN 2024

12. Name any two stations having equable climate. COMP 2021

13. Write any two parts of India which receive very low rainfall. SQP 2022

14. Give a geographical reason for each of the following : MAIN 2023

(i) Western part of Rajasthan is a desert region.

(ii) Kochi has a lesser annual range of temperature than Agra.

(iii) The Northern plains and Punjab receive winter cyclonic rain in winter.

15. Himalayas act as a climatic divide.' Justify the statement giving two reasons.
MAIN 2024

16. Name the seasons of India and mention the duration of each season.
MAIN 2023

17. Why maximum rainfall in India occurs after the summer season?
MAIN 2023

18. An Inter-School Football Match at Vishakhapatnam from November 1 to 3 had to be cancelled because of bad weather. Give an explanation for this happening. Your explanation should provide the technical name of this season in India, as well as the pressure conditions over the Bay of Bengal. MAIN 2023

19. Which of the following is NOT a source of rainfall for Punjab and Haryana?
COMP 2024

(a) Western disturbances

(b) Arabian Sea branch of South-west monsoon

- (c) Bay of Bengal branch of South-west monsoon
- (d) Retreating monsoon.

20. What causes the 'October Heat'? COMP 2022

- (a) High temperature and heavy rainfall.
- (b) Reduced temperature and heavy rainfall.
- (c) High temperature and humidity.
- (d) Reduced temperature and humidity.

**21. (i) Name two regions which get very little rain from the Summer Monsoon.
(ii) Name any two local winds which blow in India during the summer season.
SQP 2021**

22. Give a geographical reason for the following: MAIN 2024

- (i) Kolkata gets more rain than Delhi during rainy season.
- (ii) Though both are hill stations, Shimla experiences snowfall during the winters but Ooty does not.
- (iii) Winter monsoons bring less rain.

**23. (i) Name the seasons during which the North- East Trade Winds dominate.
(ii) How do these winds affect the climate of India ? COMP 2022**

24. Give a geographical reason for each of the following : MAIN 2023

- (i) The Malabar coast receives heavy rainfall in the month of July but the Coromandel coast during this time remains comparatively dry.
- (ii) Punjab receives rainfall in winter season.
- (iii) Mumbai is warmer than Delhi in the month of December.

25. Give a geographical reason for each of the following : MAIN 2022

- (i) Mumbai is warmer than Kanpur in December.
- (ii) Punjab gets rain in winter.
- (iii) The Indo - Gangetic Plain gets some rainfall in the months of December and January

26. Mention three important features of rainfall in India. MAIN 2024

27. Study the climatic data given below and answer the questions that follow:

Station	Month	J	F	M	A	M	J	J	A	S	O	N	D
A	Temperature in Degree C.	14.4	16.7	23.3	30.0	33.3	33.3	30.0	29.4	28.9	25.6	19.4	15.6
	Rainfall in cm	2.5	1.5	1.3	10	1.8	7.4	19.3	17.8	11.9	1.3	0.2	1.0
B	Temperature in Degree C.	24.4	24.4	26.7	28.3	30.0	28.9	27.2	27.2	27.2	27.8	27.2	25.0
	Rainfall in cm	0.2	0.2	0.3	10	1.8	50.6	61.0	24.8	24.8	24.8	1.0	0.7

(i) Calculate the annual rainfall for station A.

(ii) What is the annual range of temperature of station B ?

(iii) In which hemisphere do you think station A lies ?

(iv) Which of these stations has an equable climate ?

SQP 2022

28. Given below is the climatic data of a station. Study the table and answer the questions that follow:

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temperature in Degree C.	13.7	16.6	21.6	25.5	33.2	33.5	30.8	29.8	29.2	25.5	19.6	15.2
Rainfall in cms.	2.5	2.0	1.5	0.9	1.5	7.5	17.8	18.5	12.5	1.0	0.2	1.5

(i) Calculate the annual range of temperature.

(ii) What is the total rainfall experienced by the station?

(iii) Which is the driest month?

MAIN 2024

29. Study the climatic data given below and answer the questions that follow:

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temperature °C	23.1	24.8	26.5	29.3	32	32.8	33.1	32.1	30.5	29.3	28.7	26.1
Rainfall cm	15.3	10.1	0.3	0.1	1.3	4.5	6.1	10.2	10.5	20.1	16.8	19.0

- Calculate the annual rainfall experienced by the station.
- Suggest a name of this station, giving a reason for your answer.
- Name the season during which the rainfall is heaviest. MAIN 2023

30. Study the climate data of Station X and answer the questions that follow:[3]

Month	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
Temperature °C	21.0	21.9	24.3	27.2	28.0	26.4	26.1	25.4	26.0	26.0	23.8	21.2
Rainfall in cm	5.1	2.8	1.2	1.7	3.9	4.6	8.4	11.4	11.9	31.6	34.5	14.8

- Calculate the annual range of temperature of Station X.
- Name the wind that brings most of the rainfall to this station.
- On which coast of India is Station X located? MAIN 2023

Solution

1. (d) Temperate cyclone

Snowfall in Kashmir during winter is caused by **temperate cyclones**.

2. (c) Coromandel coast

The area that receives rain from the **North East Monsoon** is the **Coromandel coast**.

3. (c) Temperate Zone

The northern half of India lies in the **Temperate Zone**.

4. Monsoon is a periodic wind that changes its direction with the change in season.

5. The Monsoon wind changes its direction due to variations in temperature and pressure over the sea and land.

6. The two months associated with retreating monsoons are **October and November**.

7. (c) Patna is on eastern side, so Bay of Bengal branch brings more rain to it than to Agra.

Patna gets more rain than Agra during the summer monsoon season because it is on the eastern side, so the **Bay of Bengal branch** brings more rain to it than to Agra.

8. Rajasthan and Gujarat.

9. Leh and Jodhpur.

10. June, July and August.

11.

$$\text{Average rainfall} = \frac{\text{Total annual rainfall}}{\text{Number of months}}$$

12. Mumbai and Chennai.

13. Two parts of India which receive very low rainfall are **Rajasthan** and **some parts of Gujarat**.

14. (i) The western part of Rajasthan is a desert region because it lies in the **rain shadow region** of the Aravalli hills, receiving very little rainfall.

(ii) Kochi has a lesser annual range of temperature than Agra because it is **close to the sea**, and the **sea moderates temperature** throughout the year.

(iii) The Northern Plains and Punjab receive winter cyclonic rain in winter due to the **temperate cyclones** that form over the Mediterranean Sea and bring rainfall to these regions.

15. The Himalayas act as a climatic divide due to the following reasons:

(i) **Barrier to Cold Winds:** They block cold winds from Central Asia, keeping northern India warmer in winter.

(ii) **Monsoon Impact:** The Himalayas force the southwest monsoon winds to drop rainfall on the northern plains, while creating drier conditions on the leeward side.

16. India has four main seasons:

(i) **Winter:** December to February

(ii) **Summer:** March to May

(iii) **Monsoon:** June to September

(iv) **Post-Monsoon (Autumn):** October to November

17. Maximum rainfall in India occurs after the summer season due to the **southwest monsoon winds**. These winds, driven by the temperature difference between the Indian Ocean and the heated landmass of India, bring **moist air inland**, causing heavy rainfall, especially from June to September.

18. The Inter-School Football Match in Vishakhapatnam from November 1 to 3 was cancelled due to the **Northeast Monsoon**. During this season, **low-pressure**

conditions over the **Bay of Bengal** bring moisture-laden winds, causing heavy rainfall and storms, which disrupt outdoor activities.

19. (c) Bay of Bengal branch of South-west monsoon

The **Bay of Bengal branch of the South-west monsoon** is not a source of rainfall for Punjab and Haryana.

20. (c) High temperature and humidity

The 'October Heat' is caused by **high temperature and humidity**.

21. (i) Two regions which get very little rain from the Summer Monsoon – **Western Rajasthan and the Thar Desert**.

(ii) Two local winds that blow in India during the summer season – **Loo and Kaal Baisakhi**.

22. (i) Kolkata gets more rain than Delhi during the rainy season because it is **closer to the Bay of Bengal** and lies directly in the path of the **moisture-laden southwest monsoon winds**, whereas Delhi is further inland and receives less rainfall.

(ii) Shimla experiences snowfall in winter because it is located in the **Himalayas at a higher altitude and latitude**, causing colder temperatures, while Ooty, in the **lower Nilgiri Hills near the equator**, has milder temperatures.

(iii) Winter monsoons bring less rain because the winds blow from the **land to the sea**, carrying **dry continental air** with very little moisture.

23. (i) The **North-East Trade Winds** dominate during the **winter season (December to February)**, also called the **retreating monsoon**.

(ii) These winds affect the climate of India by bringing **cooler and dry weather** to most parts of the country, but they cause **rainfall along the southeastern coast** (Coromandel coast) due to moisture picked up from the Bay of Bengal.

24. (i) The Malabar coast receives heavy rainfall in July because it lies on the **windward side of the Western Ghats**, so the southwest monsoon winds drop their moisture here. The Coromandel coast is on the **leeward side**, so it remains comparatively dry.

(ii) Punjab receives rainfall in the winter season due to **western disturbances**, which are temperate cyclones originating from the Mediterranean region.

(iii) Mumbai is warmer than Delhi in December because it is **close to the sea**, and the **sea moderates the temperature**, whereas Delhi, being inland, experiences more extreme cold.

25. (i) Mumbai is warmer than Kanpur in December because it is **near the sea**, and the **sea moderates temperatures**, whereas Kanpur, being inland, experiences colder winter temperatures.

(ii) Punjab gets rain in winter due to **western disturbances**, which are temperate cyclones originating from the Mediterranean region.

(iii) The Indo-Gangetic Plain gets some rainfall in December and January because **western disturbances** bring **winter rain** to this region.

26. Three important features of rainfall in India are:

(i) **Concentrated Monsoon Rainfall:** About **70% of annual rainfall** occurs during the monsoon season (June to September), while the rest of the year is largely dry.

(ii) **Non-Monsoon Rainfall:** A small portion of rainfall comes from **cyclonic and convectional sources**.

(iii) **Erratic Nature:** The **quantity and timing of rainfall are unpredictable**, leading to floods or droughts in different regions.

27. (i) The annual rainfall of station A is 67.0 cm.

(ii) Annual range of temperature of station B is $30^{\circ}\text{C} - 24^{\circ}\text{C} = 6^{\circ}\text{C}$.

(iii) Northern Hemisphere.

(iv) Station B.

28. (i) Annual range of temperature: $(33.5 - 13.7)^{\circ}\text{C} = 19.8^{\circ}\text{C}$.

(ii) 67.4 cm

(iii) November is the driest month.

29. (i) The annual rainfall experienced by the station is $(15.3 + 10.1 + 0.3 + 0.1 + 1.3 + 4.5 + 6.1 + 10.2 + 10.5 + 20.1 + 16.8 + 19.0) = 114.3$ cms.

(ii) The station is called Chennai due to its high rainfall from October to January.

(iii) The winter months experience the highest levels of rainfall.

30. (i) Annual range of temperature = $(28.0 - 21.0)^{\circ}\text{C} = 7^{\circ}\text{C}$

(ii) The Retreating Monsoons

(iii) The station X lies on the eastern coast of South India.